

AMENDMENT NO. 2 MARCH 2025
TO
IS 3224 : 2021 VALVE FOR COMPRESSED GAS CYLINDERS EXCLUDING
LIQUEFIED PETROLEUM GAS (LPG) CYLINDERS — SPECIFICATION
(Fourth Revision)

(Page 7, clause 5.1) — Insert the following at the end:

‘5.2 Chemical Composition

Metallic and non-metallic materials in contact with the gas shall be compatible with the gas, according to IS/ISO 11114-1 and IS/ISO 11114-2 under all intended operating conditions. For valves used for dissolved gases, the compatibility of the materials in contact with the solvent shall also be considered. For valves used with gas mixtures, the compatibility of the gas wetted materials with each component of the gas mixture shall be considered.

When using plated or coated components in gas wetted areas the material compatibility of both, the plating/coating material and the substrate material shall be taken into account. In addition, consideration should be given to avoid flaking or particle generation, especially for oxygen, other oxidizing gases and gas mixtures containing oxygen or other oxidizing gases.

The actual chemical composition shall be the subject of agreement between the purchaser and the manufacturer except for chlorine and boron trifluoride valves.

5.2.1 Material of construction for valve of chlorine and boron trifluoride cylinders shall be as under.

5.2.1.1 *Valve body*

5.2.1.1.1 *Aluminum silicon bronze*

Chemical composition:

Copper : Remainder	Lead : 0.05 percent, <i>Max</i>	Tin : 0.20 percent, <i>Max</i>
Iron : 0.30 percent, <i>Max</i>	Nickel : 0.25 percent, <i>Max</i>	Aluminum : 6.3 to 7.0 percent
Silicon : 1.5 to 2.0 percent	Manganese : 0.10 percent, <i>Max</i>	Zinc : 0.50 percent, <i>Max</i>
Arsenic : 0.15 percent, <i>Max</i>	—	—

NOTE — For guidance IS 6912 designation FABS may be referred.

Valve body forging shall be heat treated between 580 °C and 607 °C for 4 h and to be air cooled in open atmosphere.

5.2.1.2 *Valve spindle*

- a) Valve spindle shall be machined from wrought nickel-copper alloy;
- b) Typical composition of the alloy including admissible impurities given below are for guidance only;
and

Ni (+Cobalt): 63 percent, *Min*;

Cu: 27 percent to 34 percent;

Fe: 1.0 percent to 2.5percent

Price Group 1

Amendment No. 2 to IS 3224 : 2021

- c) Maximum admissible impurities/alloying elements: As agreed to between the purchaser and the manufacturer.

5.2.1.3 *Other extruded brass components*

Other extruded brass components for example gland nut, packing gland, packing collar, outlet cap etc.; are generally made from free cutting brass rods (*see* IS 319), naval brass, forging quality brass such as lead brass conforming to IS 6912 or any material as agreed to be between the manufacturer and the purchaser.

5.2.2 Copper alloys in contact with oxygen or other oxidizing gases or gas mixtures shall not have aluminium content more than 2.5 percent.

5.2.3 Nickel shall not be used in gas-wetted areas with carbon monoxide due to formation of nickel carbonyl.'

(MED 16)

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